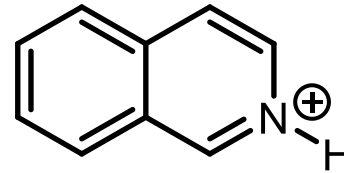
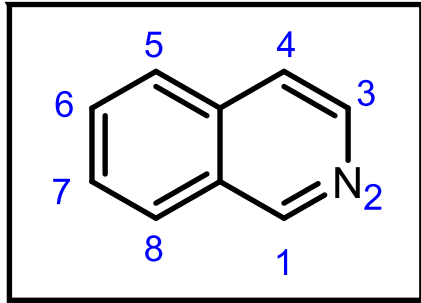
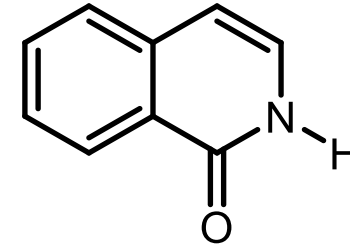


5.1.3. Isochinolin (engl. isoquinoline)

5.1.3.1. Struktur, Eigenschaften, Vorkommen, Bedeutung



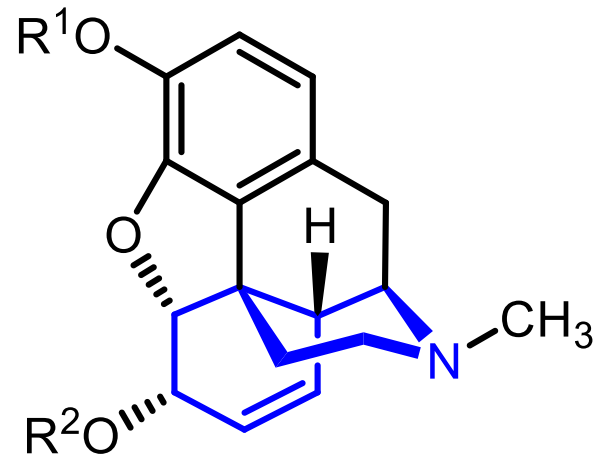
Isochinolinium-Ion



1-Isochinolon

Farblose Substanz mit angenehmem Geruch; Schmp. 26 °C, Sdp.: 243 °C

Isochinolinalkaloiden, z. B. Morphin, Codein (in Hustensäften), Heroin

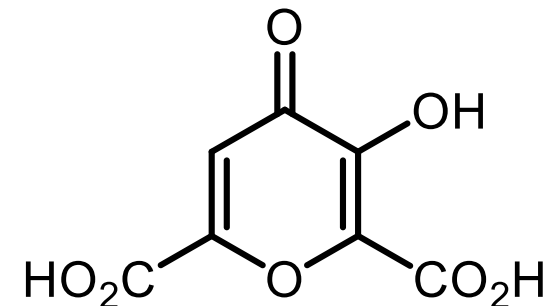


Morphin (R¹, R² = H)

Codein (R¹ = CH₃, R² = H)

Heroin (R¹, R² = Ac)

Alkaloide oft als Salze
der Mekonsäure



5.1.3.1. Struktur, Eigenschaften, Vorkommen, Bedeutung (Isochinolin) *Morphine*

Gewinnung: Rohopium durch abendliches horizontales oder vertikales Anritzen der grünen Kapseln 8-14 Tage nach dem Abfallen der Kronblätter und Abkratzen des ausgetretenen, erhärteten und braun verfärbten Milchsafts am nächsten Morgen (20.000 Mohnkapseln liefern 1 kg Opium). Zusammensetzung von Rohopium: zu 20-30 % bestehend aus ca. 40 (50) verschiedenen, überwiegend als Salze der Mekonsäure sowie der Milchsäure, Fumarsäure und Schwefelsäure vorliegenden Alkaloiden vom Morphinan- und Benzylisochinolin-Typ; vom Morphinan-Typ insbesondere Morphin (3-23 %), Codein (0.2-3.5 %) und Thebain (0.2-1.0 %), vom Benzylisochinolin-Typ Papaverin (0.5-3.0 %), Noscapin (= Narcotin, 2-12 %) und Narcein (0.1-0.7 %).



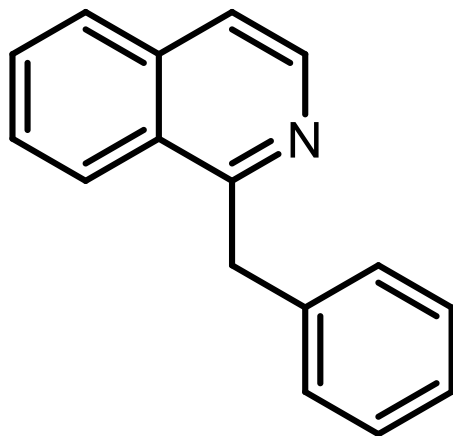
Papaver somniferum

Morphinwirkungen:

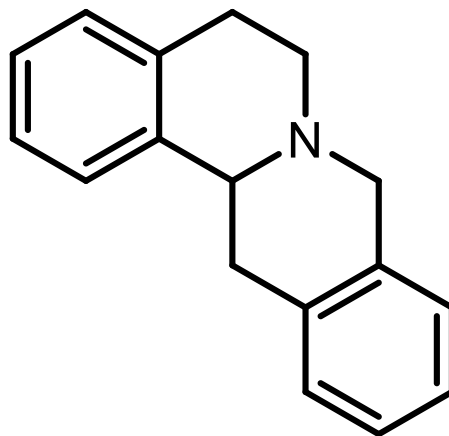
zentral wirksames Analgetikum durch Angriff an Opioid-Rezeptoren (analgetisch wirksame Dosis 10 mg, sedativ-hypnotische Wirkung bei einer Dosierung von 10 mg, narkotische Wirkung bei Dosierung von 50-100 mg).

5.1.3.1. Struktur, Eigenschaften, Vorkommen, Bedeutung (Isochinolin)

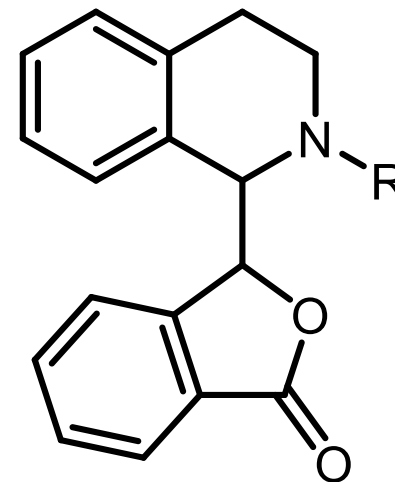
Isochinolin-Alkaloide



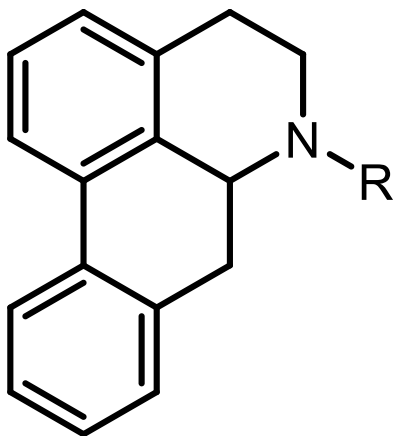
Benzyloisochinolin-Typ



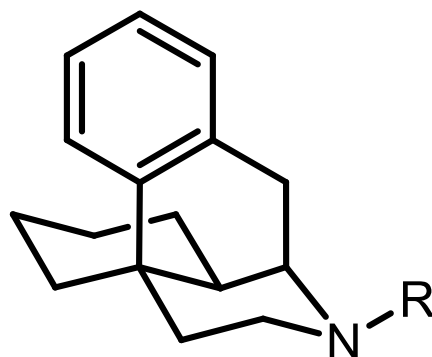
Protoberberin-Typ



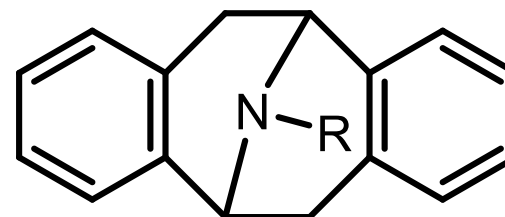
Phthalide-Isoquinolin-Typ



Aporphin-Typ



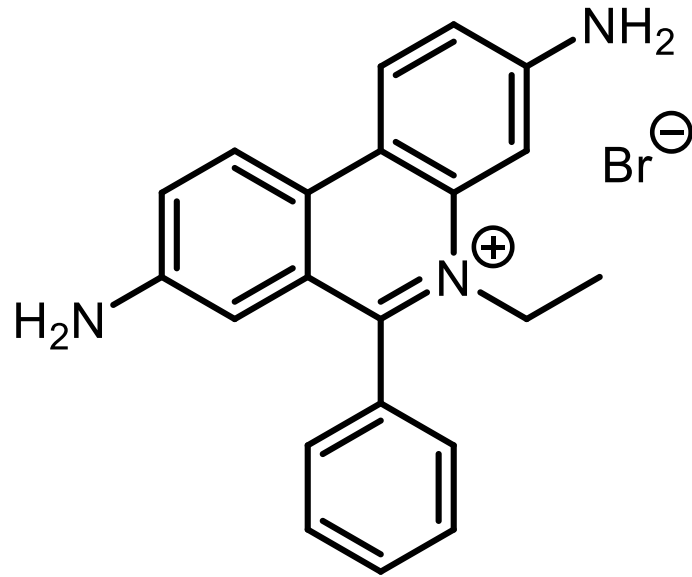
Morphinan-Typ



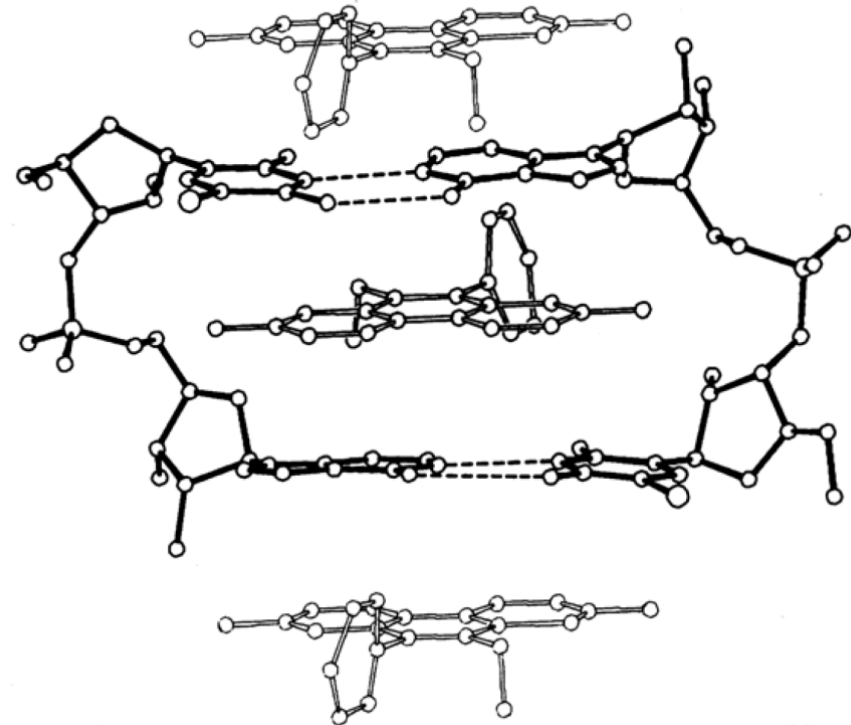
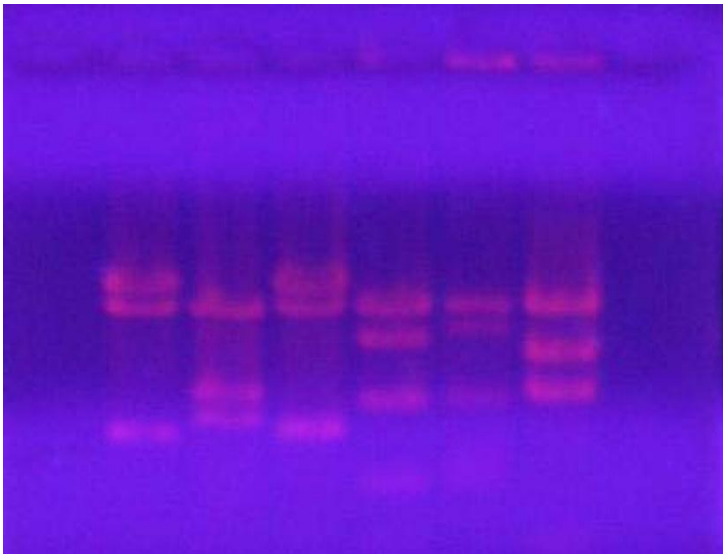
Pavin-Typ

5.1.3.1. Struktur, Eigenschaften, Vorkommen, Bedeutung (Isochinolin)

Ethidiumbromid (3,8-Diamino-5-ethyl-6-phenyl-phenanthridiniumbromid)



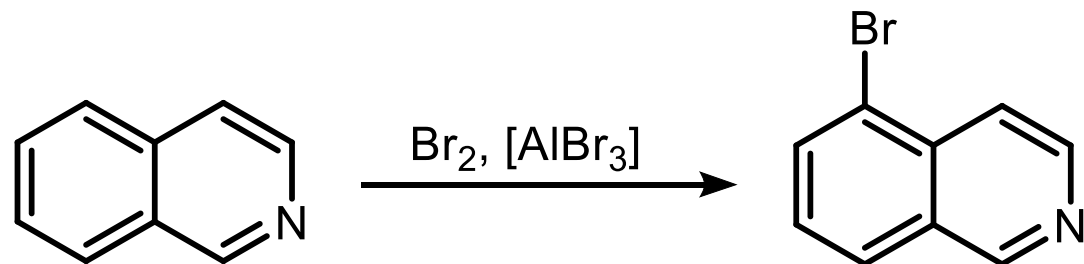
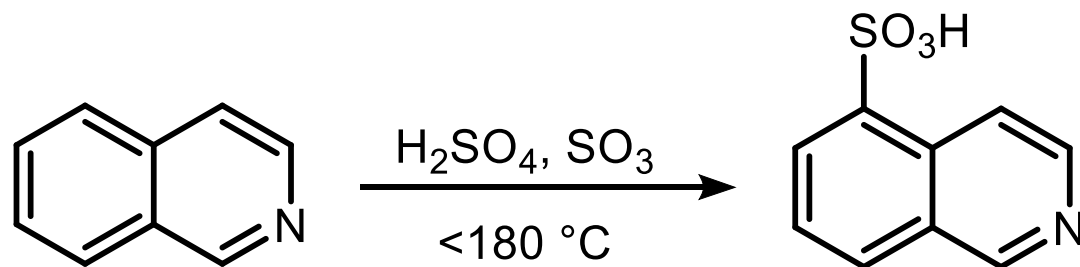
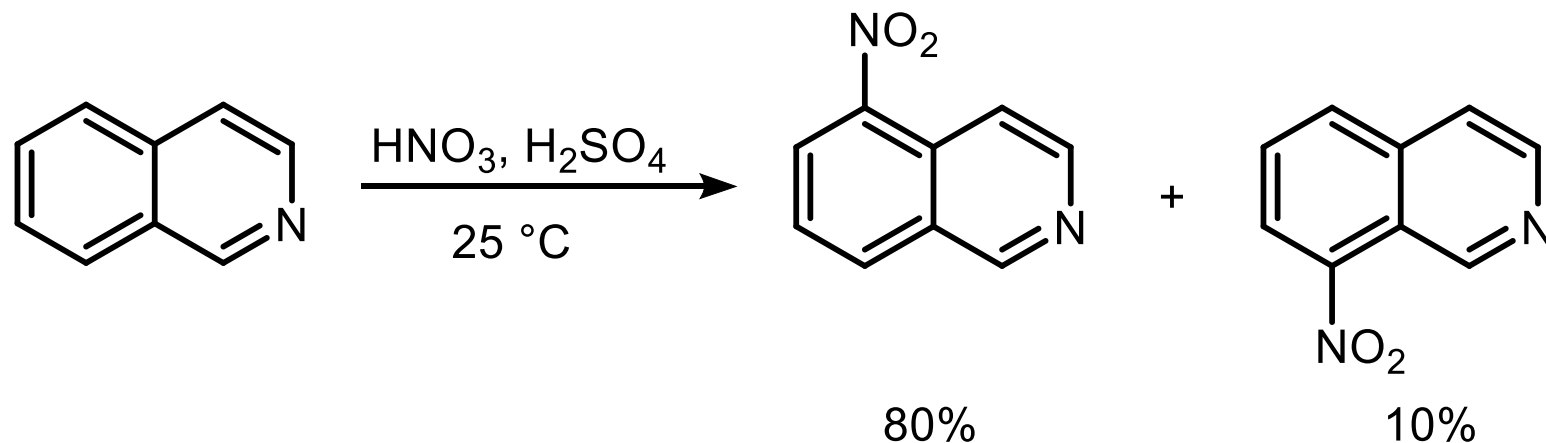
Quantifizierung von DNA und RNA nach der Gelelektrophorese (bis zu 3 Moleküle je 10 Basen) auf Basis der Erhöhung der Fluoreszenz der Verbindung nach Interkalation (erstmalig 1973).



Struktur des Ethidium-(5-Ioduridylyl(3'-5')-Adenosin)-Komplexes im Kristall (Sobell et al., *PNAS* **1975**, 72, 628).

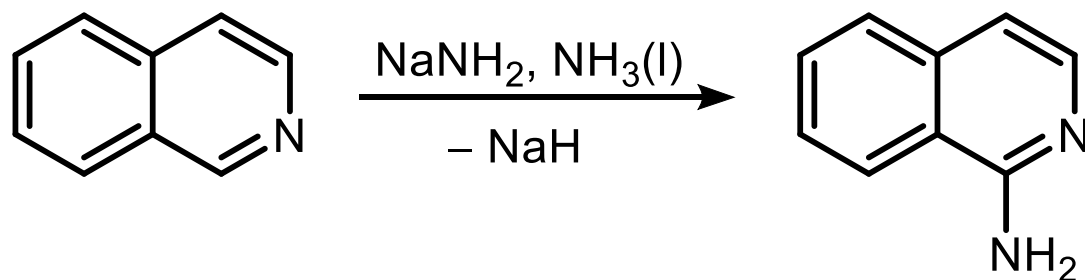
5.1.3.2. Reaktionen

1) Elektrophile Substitutionsreaktionen (S_EAr):

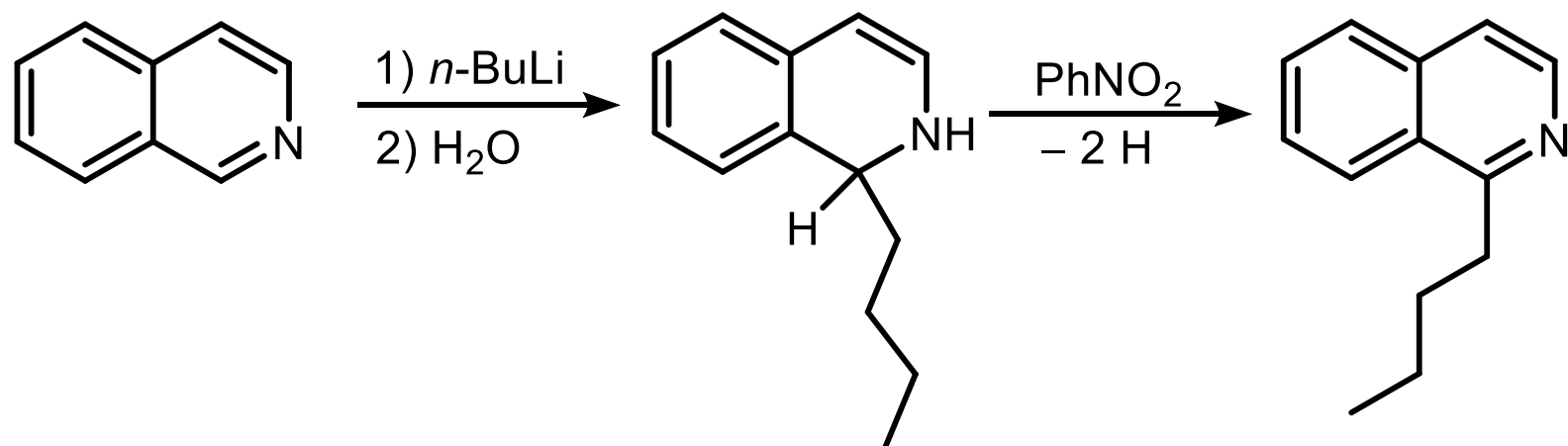


2) Nucleophile Substitutionsreaktionen S_NAr :

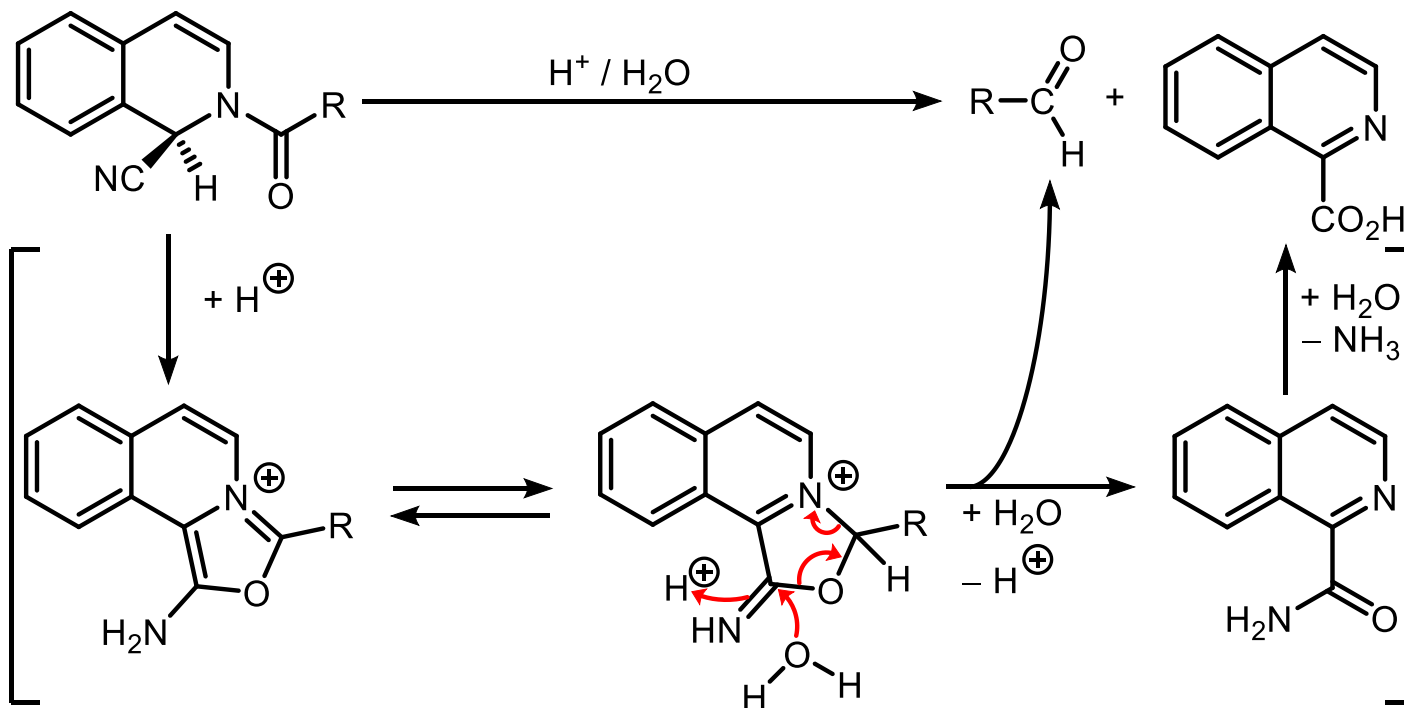
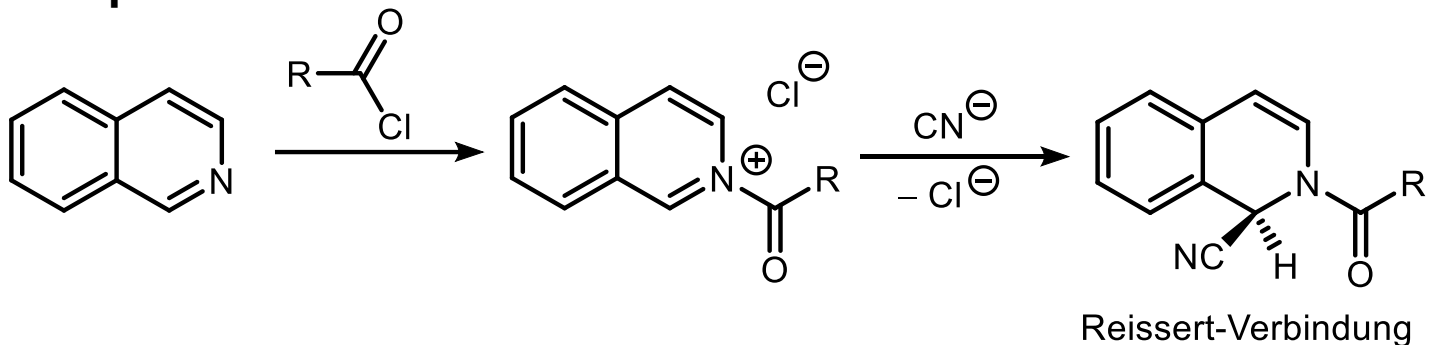
Tschitschibabin-Reaktion



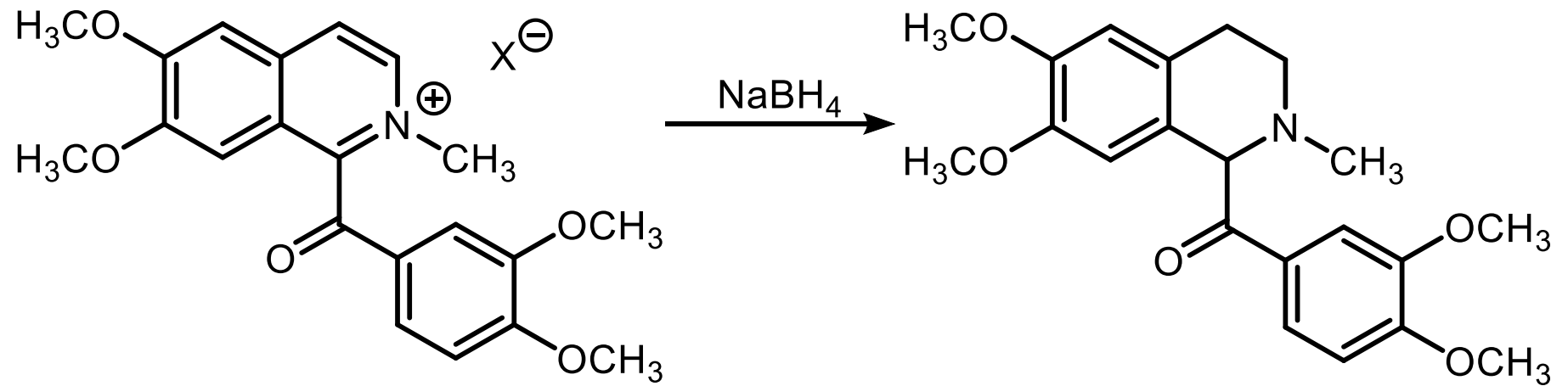
Ziegler-Reaktion



3) Nucleophile Addition: Reissert-Reaktion

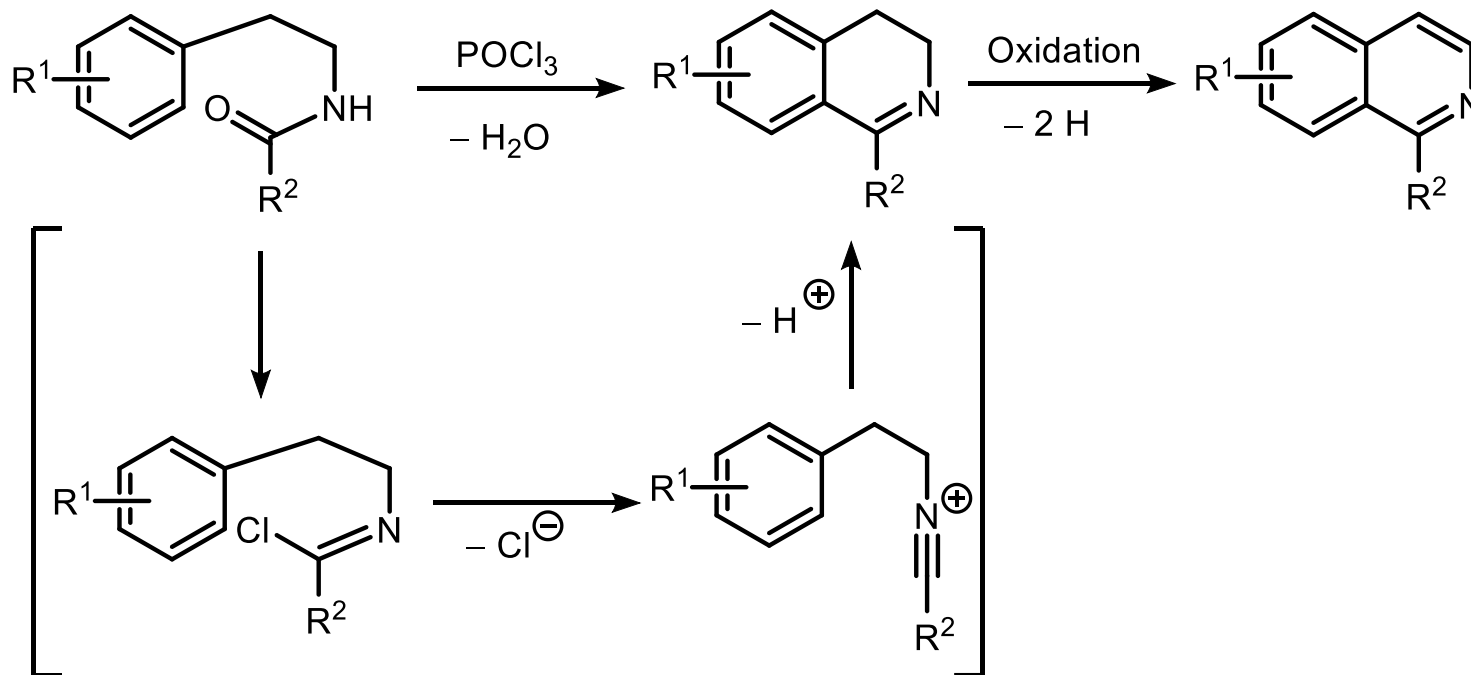


4) Reduktion

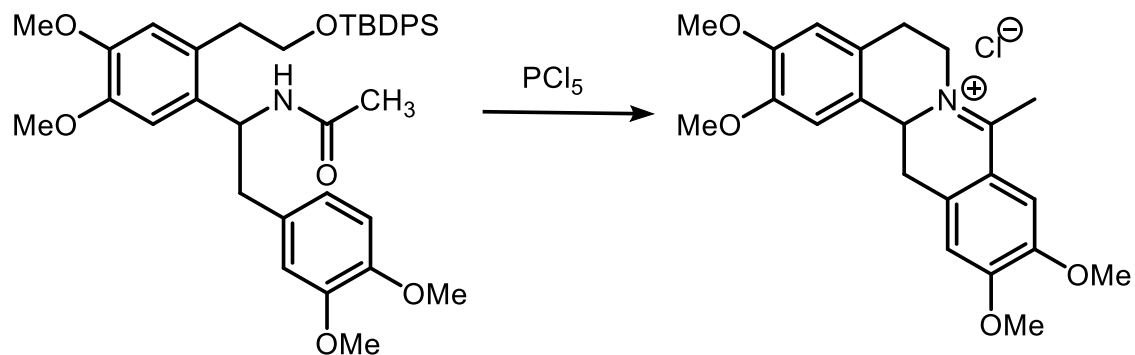


5.1.3.3. Ringaufbauende Synthesen

1) Bischler-Napieralski-Synthese



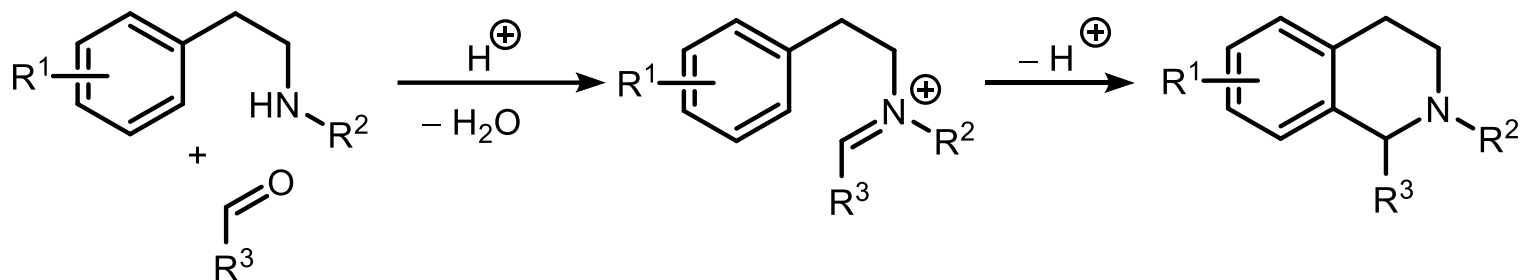
N. Sotomayor, E. Dominguez, E. Lete,
J. Org. Chem. **1996**, *61*, 4062.



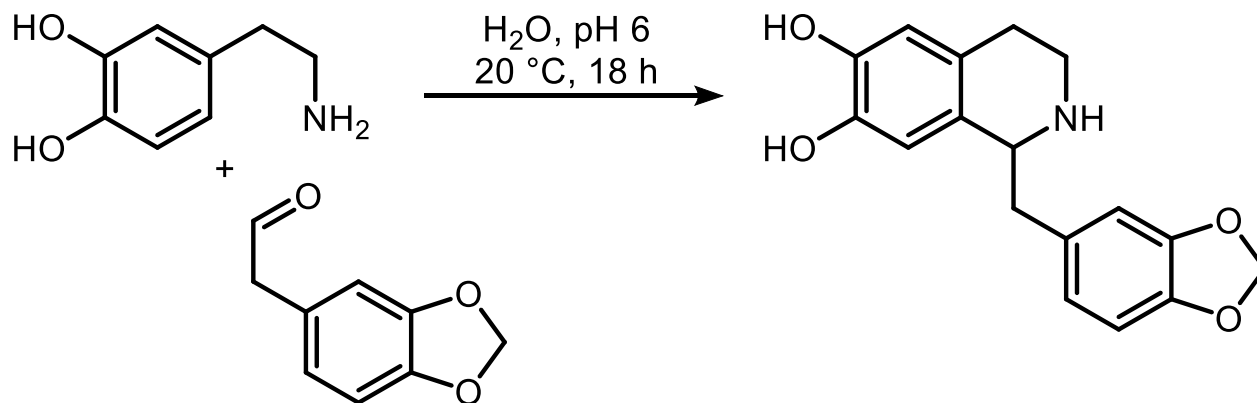
TBDPS: *tert*-Butyl-diphenylsilyl
 Prof. Dr. T. J. J. Müller

2) Pictet-Spengler-Synthese

89



Bedingungen ähnlich der Mannich-Reaktion (biomimetisch)



The image displays three chemical reaction schemes for the synthesis of quinoline derivatives.

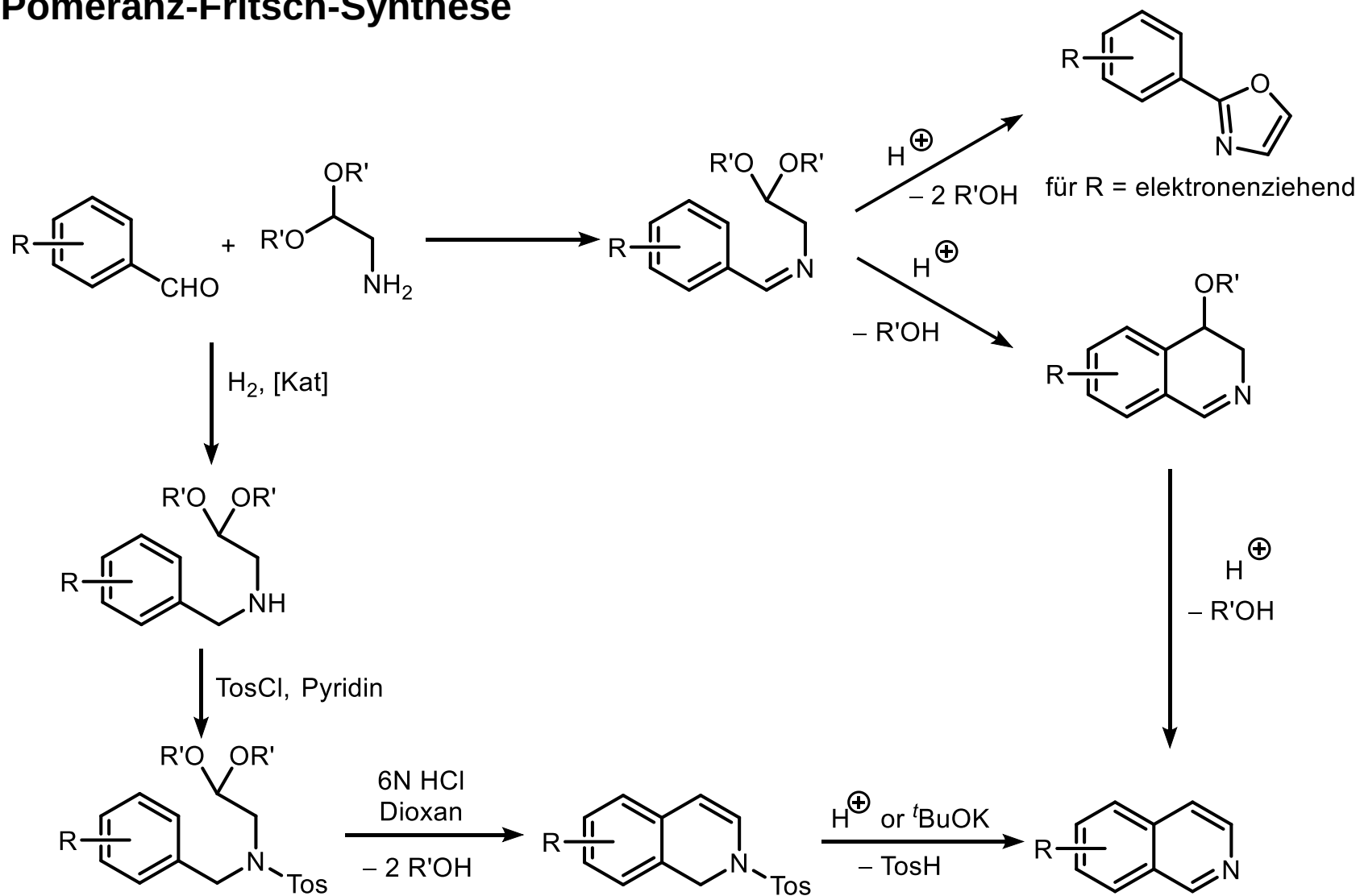
Top Scheme: A central benzophenone derivative (1-phenylpropan-1-one) reacts with three different reagents to form quinoline derivatives:

- Reaction with NH_3 yields Quinoline.
- Reaction with NH_2OH yields Quinoline-2-oxide.
- Reaction with RNH_2 and HX yields a Quinoline N-oxide salt (Quinoline-2-oxide N-oxide).
- Reaction with RNHNH_2 yields a Quinoline N-oxide derivative.

Middle Scheme: A benzofuran derivative (2-(dimethylamino)-2-oxo-1-phenylethan-1-one) reacts with PhCH_2NH_2 to form a quinoline derivative (2-(benzylamino)-2-oxo-1-phenylethan-1-one), with the loss of H_2O and Me_2NH .

Bottom Scheme: A 4-methyl-2-cyano-1-phenylethan-1-one derivative reacts with HCl to form a quinoline derivative (4-methyl-2-chloroquinoline-3-ol), with the loss of HCl .

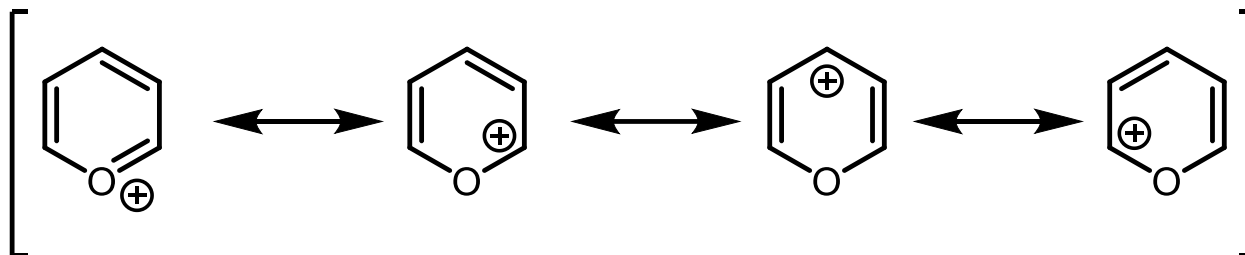
4) Pomeranz-Fritsch-Synthese



5.1.4. Pirylium-Salze

5.1.4.1. Struktur

90

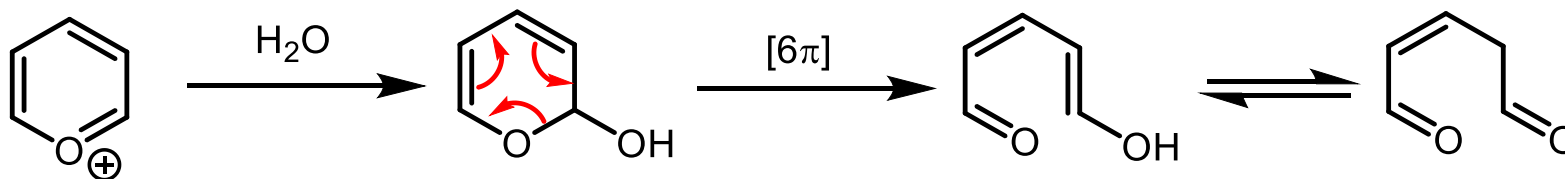


=> Angriff durch Nucleophile in 2/4- und 6-Stellung.

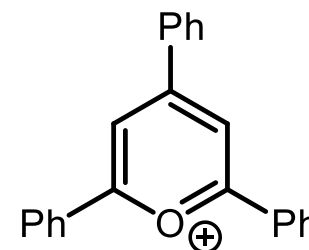
Aromatisch, aber sehr reaktiv, weil Oxoniumion-Charakter; Perchlorat: stabil bis 275°C, reagiert heftig mit Wasser!

5.1.4.2. Reaktionen

Reaktion mit Wasser



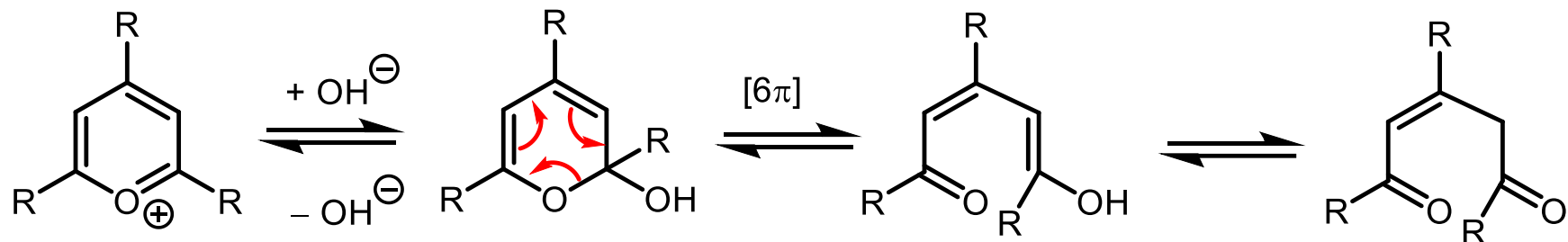
mit Substituenten wesentlich stabiler und unreaktiver!



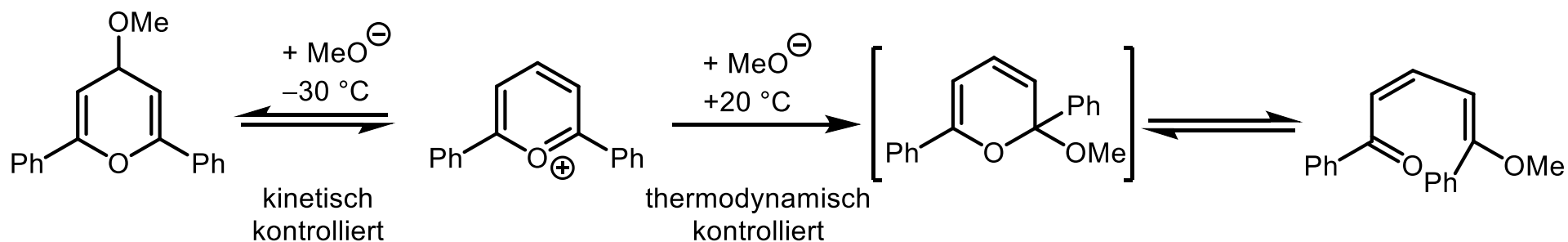
5.1.4.2. Reaktionen

Hydrolyse

Reversible Hydrolyse:



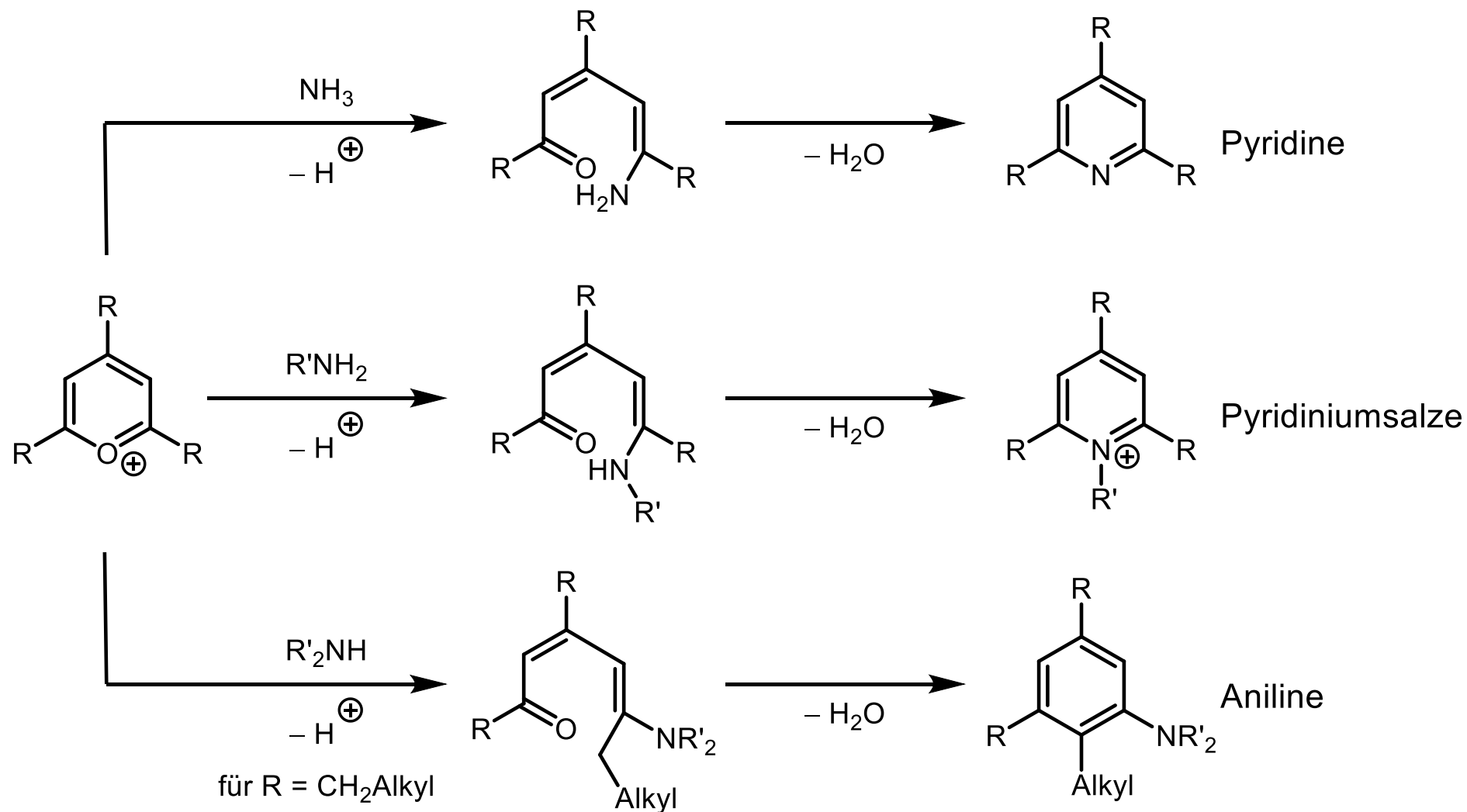
Kinetische/thermodynamische Kontrolle:



sterische Hinderung
des Angriffs in 2-Position

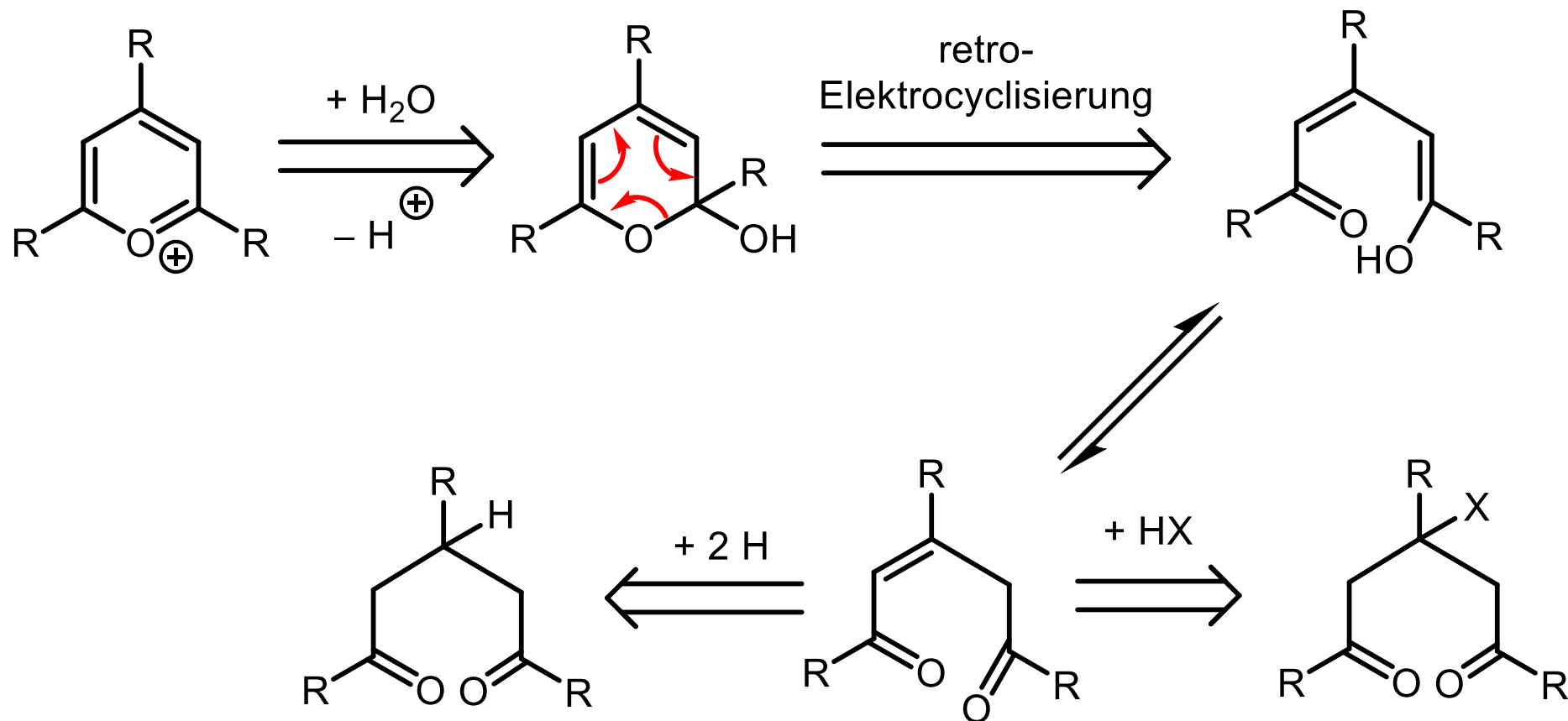
5.1.4.2. Reaktionen

Nucleophile Aminaddition



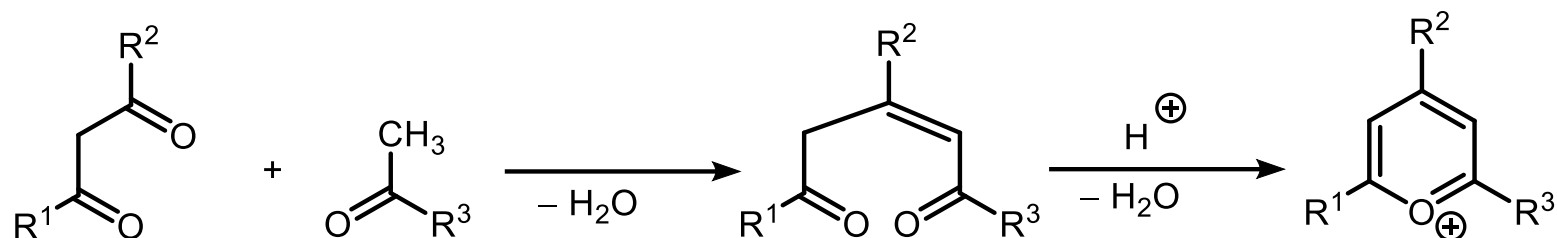
5.1.4.3. Ringaufbauende Synthesen

Retrosynthetische Analyse

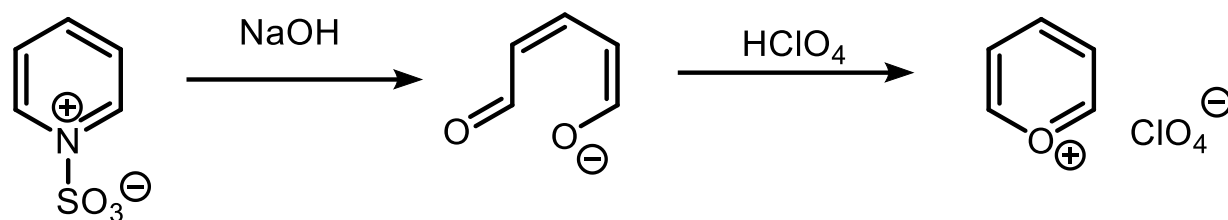


5.1.4.3. Ringaufbauende Synthesen

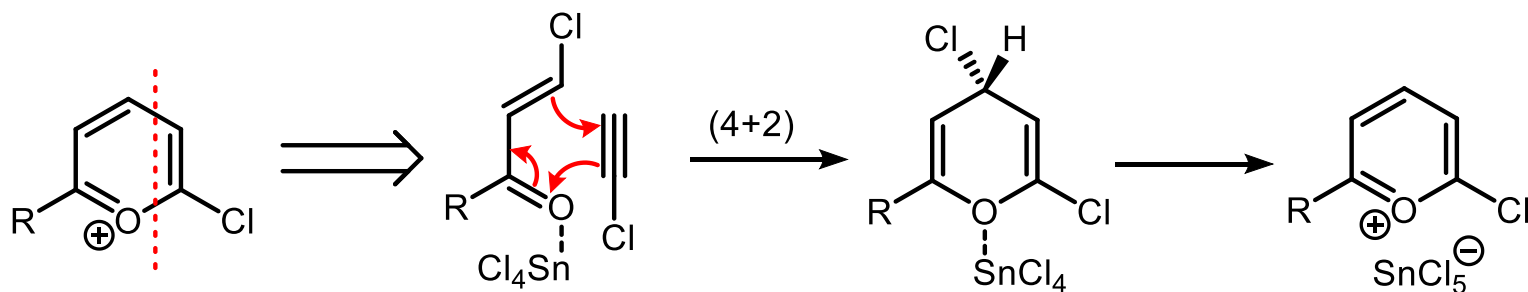
Aus 1,3-Dicarbonylverbindungen und aromatischen Methylketonen:



Aus Pyridiniumionen durch nucleophile Spaltung und saure Cyclisierung:

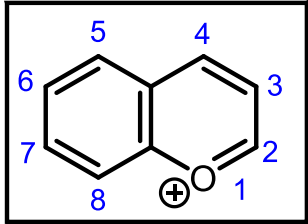


Aus β-Chlorenonen und Chloracetylenen:

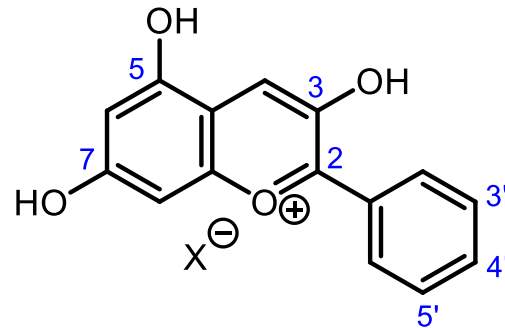


5.1.5. Benzopyrylium-Salze

5.1.5.1. Struktur, Vorkommen



Vorkommen



Flavyliumsalze

als Glycosid: Anthocyane

Aglykon: Flavyliumsalze (Anthocyanidine)

4'-OH: Pelargonidin

3',4'-OH: Cyanidin

3',4',5'-OH: Delphinidin

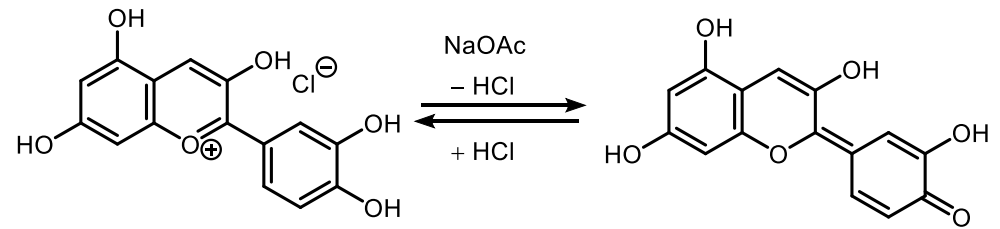


Herbstlaub



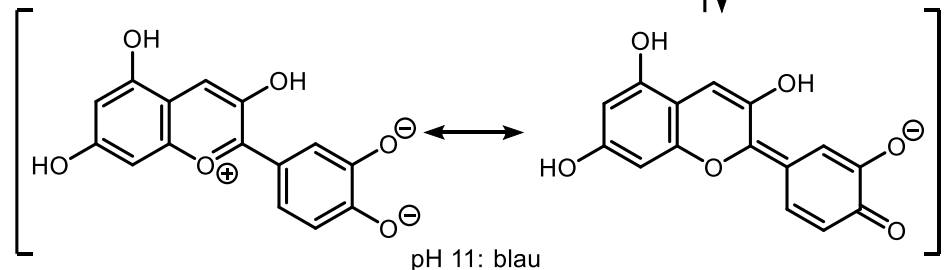
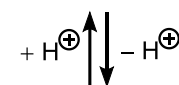
Kornblume
(*centaurea cyanus*)
(Fe²⁺ - Komplex
des Catechols)

Cyanidinchlorid



pH < 3: rot

violett

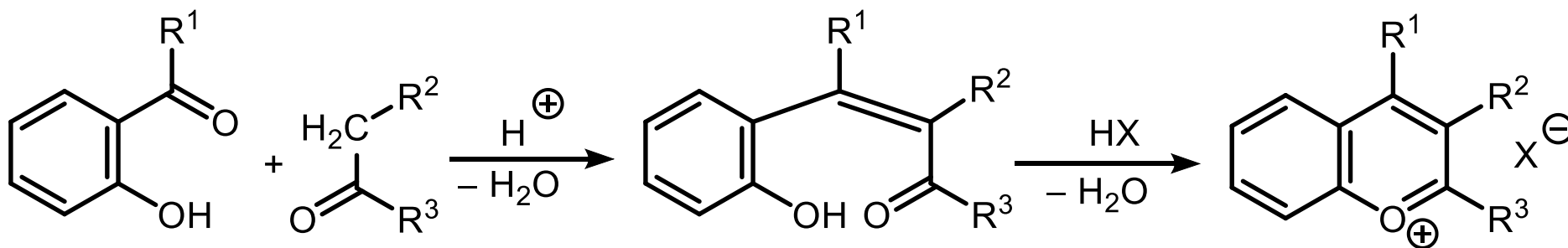


pH 11: blau

Heterocyclen

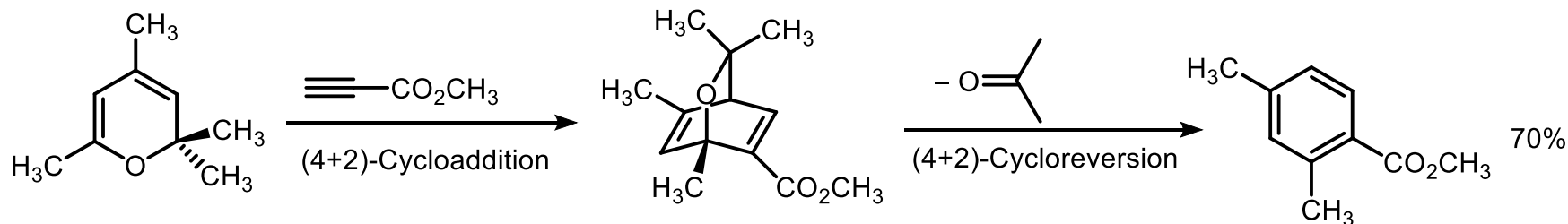
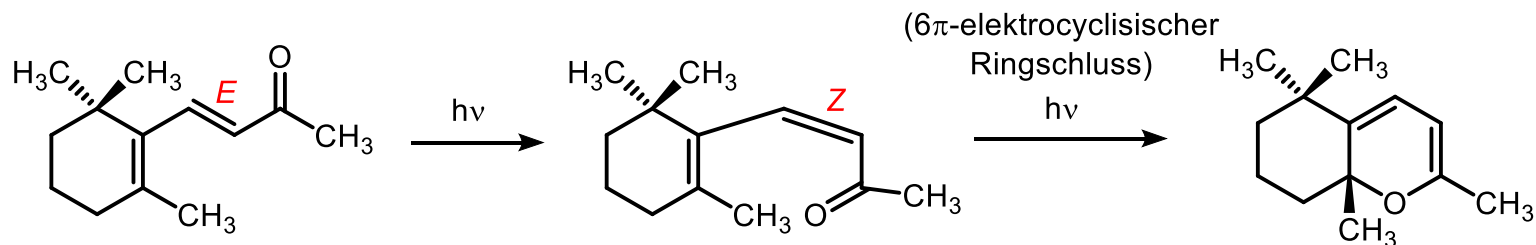
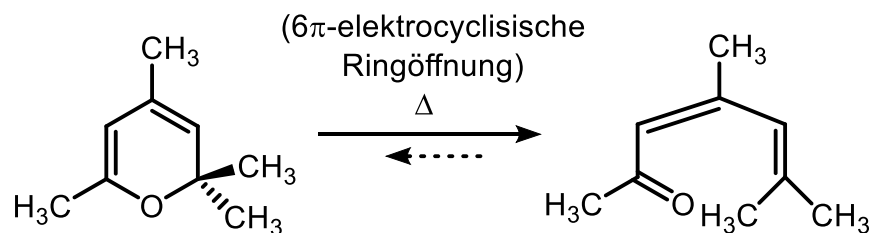
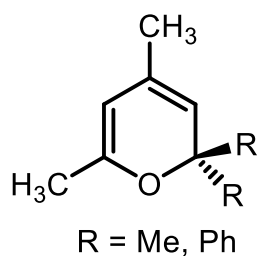
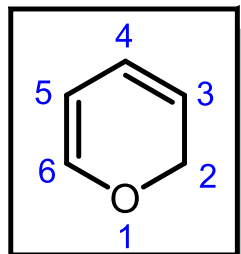
5.1.5.2. Ringaufbauende Synthesen

Cyclokondensation von o-Hydroxybenzaldehyden oder -acetophenonen mit Methylenketonen:

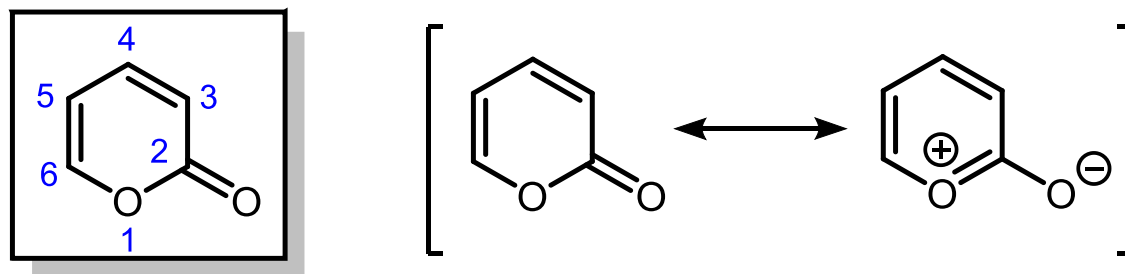


5.1.6. 2H-Pyran und Derivate

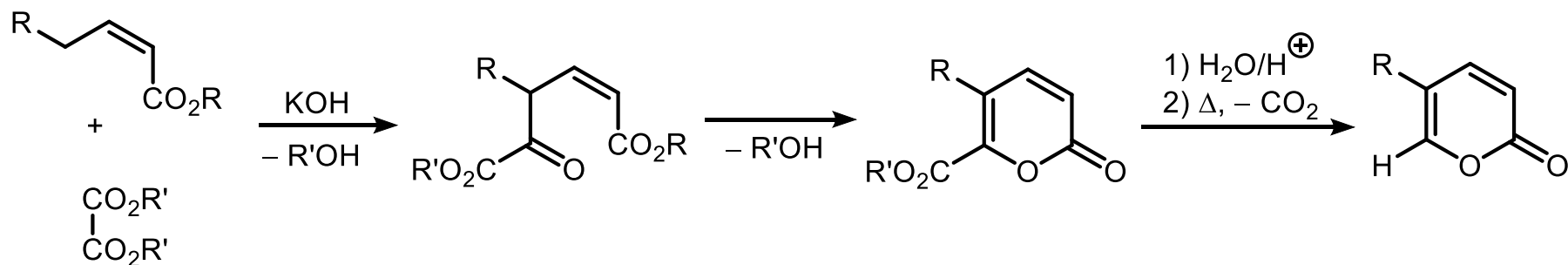
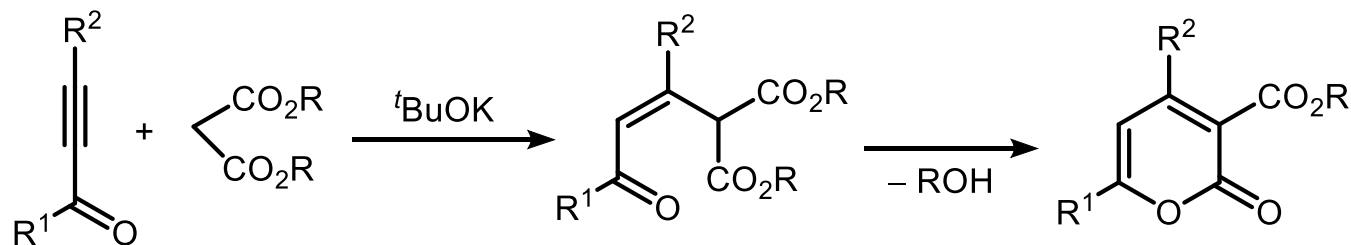
5.1.6.1. Stammsystem



5.1.6.2. 2H-Pyranon

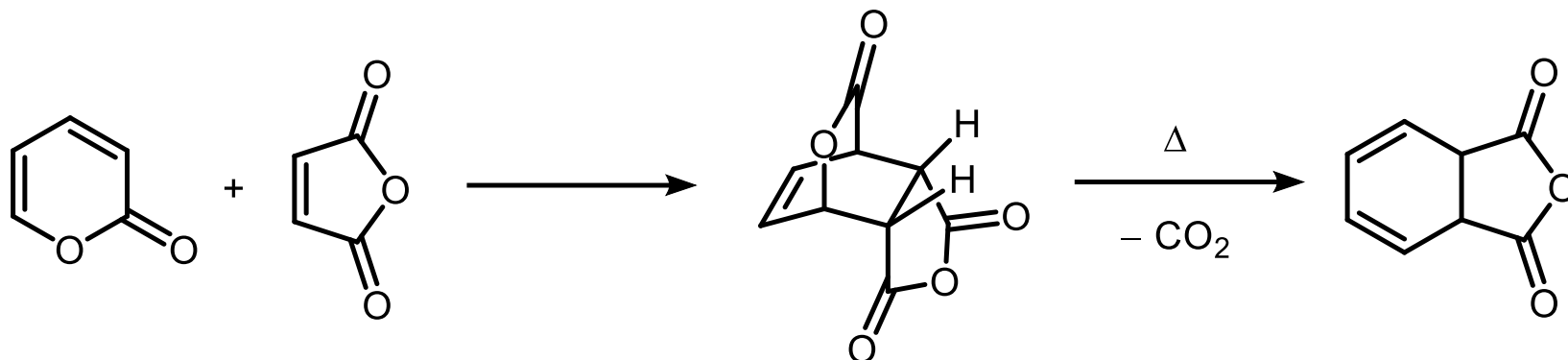


Synthesen

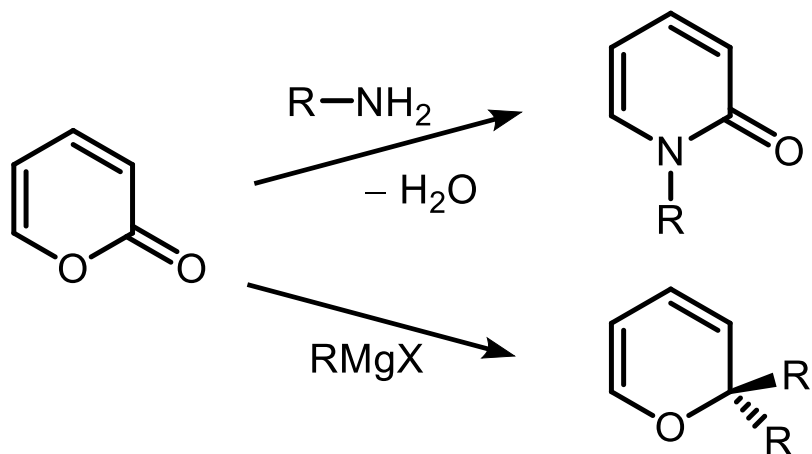


Reaktionen

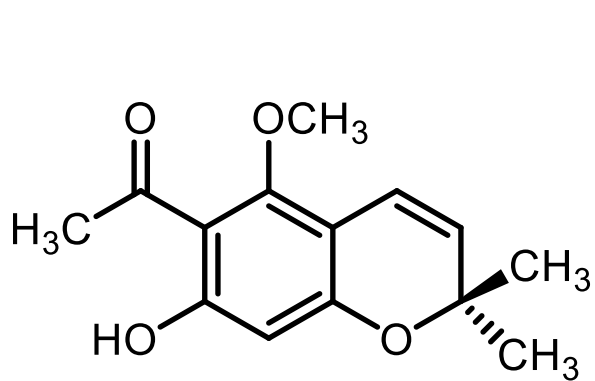
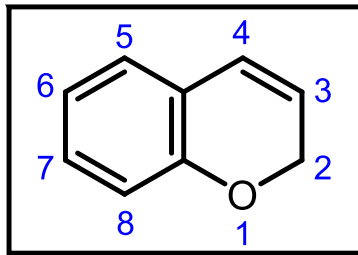
1) Diels-Alder-Reaktionen



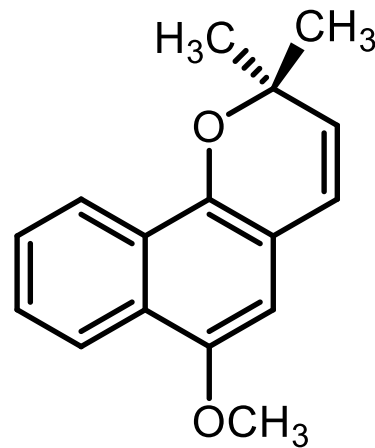
2) Angriff von Nucleophilen (an C2 der Lactonfunktion)



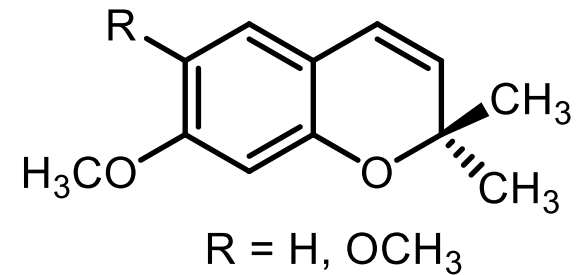
5.1.6.3. 2H-Chromen



Evodionol



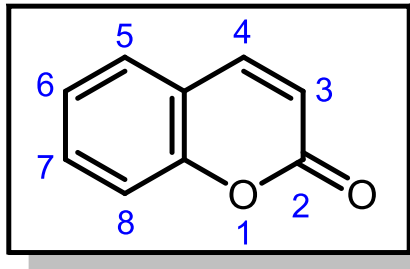
Lapachenol



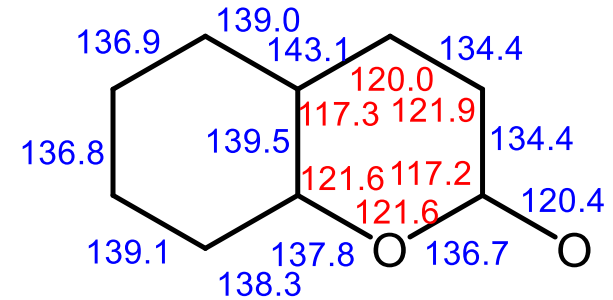
$\text{R} = \text{H}, \text{OCH}_3$

Precocen I/II

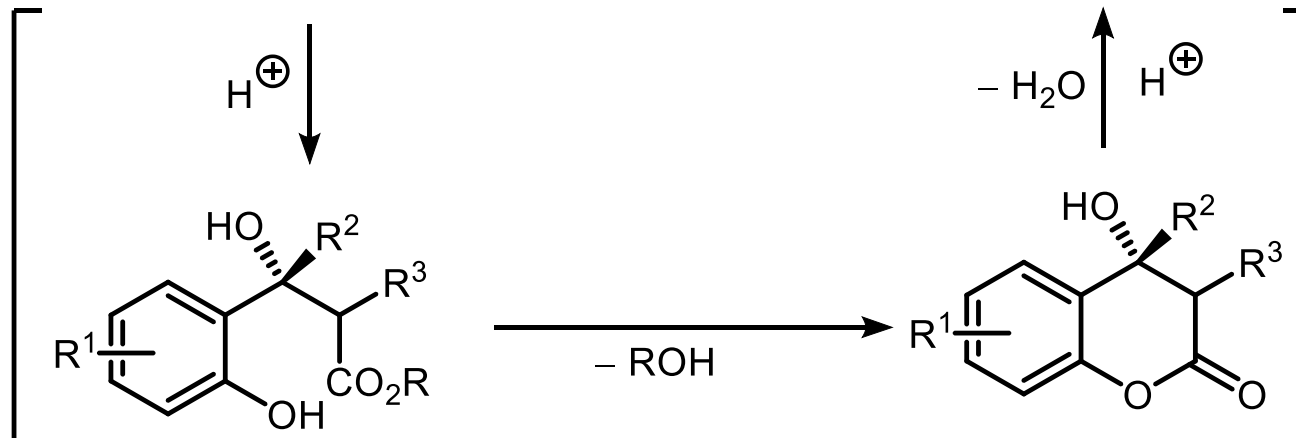
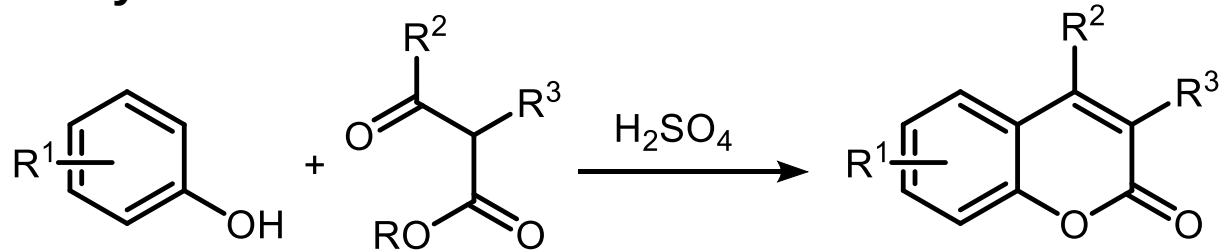
5.1.6.4. 2*H*-Chromen-2-on (Cumarin)



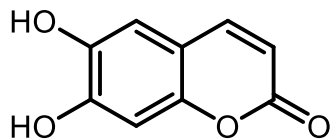
Bindungslängen [pm]
Bindungswinkel [°]



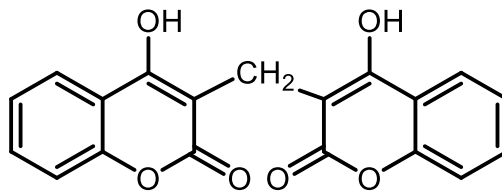
Cumarine werden oft als Laserfarbstoffe verwendet
von Pechmann-Synthese



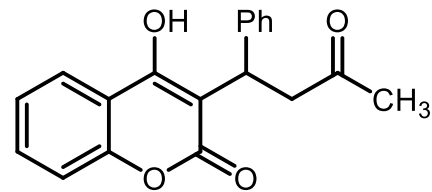
5.1.6.4. 2H-Chromen-2-on (Cumarin)



Aesculetin
(aus Roßkastanienrinde)

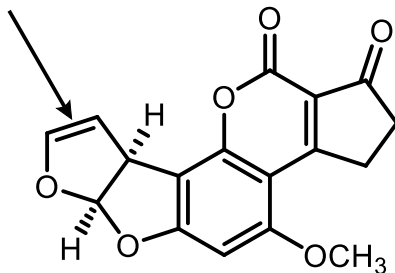


Dicumarol
(Thrombose-Behandlung)



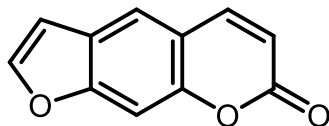
Warfarin

- 1) Epoxidierung,
- 2) Guanin-N7-Alkylierung (DNA)



Aflatoxin B1
(Schimmelpilze)

Mycotoxin Aflatoxin B1 aus
Aspergillus flavus
(lethale Dosis 1-10 mg/kg, "Fluch der
Pharaonen", Schimmelpilz bei Nüssen,
Mais, Mehl)

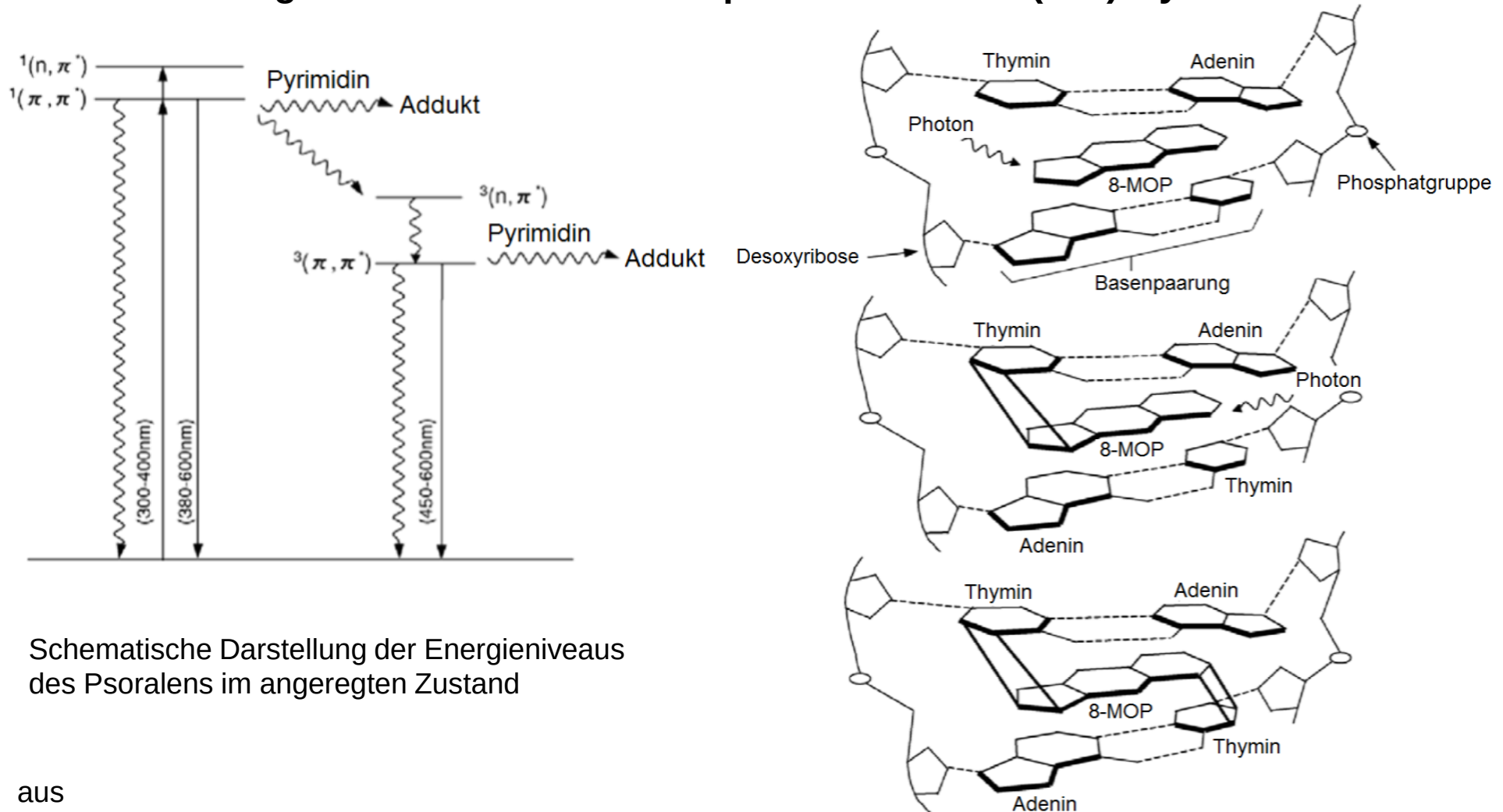


Psoralen

(aus der indischen Pflanze *Psoralea corylifolia*,
phototoxisches Furocumarin)

Verwendung von Psoralen z. B. zur
Photochemotherapie gegen Psoriasis
(Schuppenflechte).

Wechselwirkung von Psoralen mit DNA: photochemische (2+2)-Cycloaddition



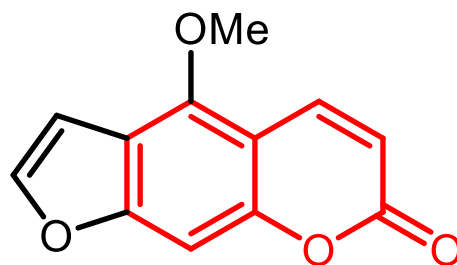
5.1.6.4. 2H-Chromen-2-on (Cumarin)



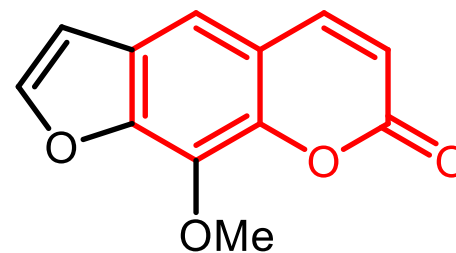
Riesenbärenklau (*Heracleum mantegazzianum*, > 3 m)

Riesenbärenklau ist giftig durch: Bergapten, Xanthotoxin, Pimpinellin, die nach Bestrahlung benetzter Haut mit Sonnenlicht Dermatitis auslösen.

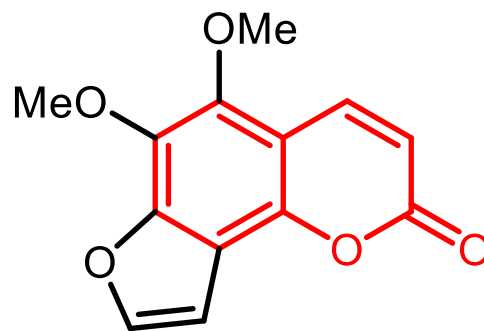
Mech. der Phototoxizität: s. Eriksson et al., *J. Photochem. Photobiol.* **2003**, 154. 235.



Bergapten



Xanthotoxin



Pimpinellin